OCTAL

Car Price Prediction System

Software Requirements Specification

Vehicle Price Prediction & Market Insights Platform

OCTAL – Car Price Prediction System

Software Requirements Specifications

7/2026

Software Requirements Specifications

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OCTAL: Car Price Prediction – Octal Team

Software Requirements Specifications

Team

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- **This document is the Software Requirements Specification (SRS) for OCTAL, a web-based Car Price Prediction system that estimates the market price of a used car from its specifications, offers a prediction path based on VIN alone, and presents market insights through an embedded dashboard.**
- **Target audience: individual car buyers and sellers looking for a fair price estimate; insurance companies, for an initial/preliminary vehicle valuation; and the development team and project supervisors, as a reference for the system's requirements.**

Introduction

Software Purpose

- **OCTAL predicts the estimated market price of a used car based on its specifications, either through manual data entry or from a VIN alone, and presents supporting market insights through an embedded analytics dashboard helping buyers, sellers, and insurance companies reach a fair price estimate.**

Software Scope

- Angular web frontend consuming a FastAPI backend that serves a trained machine learning model
- Price prediction from manually entered car specifications
- Price prediction from a VIN alone (the system decodes the vehicle's specifications automatically)
- An embedded market dashboard (Tableau) built on the project's EDA of the vehicle dataset
- An admin logs page listing all prediction queries made through the system

Definitions, acronyms, and abbreviations

- VIN: Vehicle Identification Number, a unique code identifying a specific vehicle
- EDA: Exploratory Data Analysis, the process of analyzing the vehicle dataset to understand its patterns and relationships

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- API: Application Programming Interface, used by the frontend to communicate with the backend

Requirements

Functional Requirements

3.1 Price Prediction by Manual Input

Description: The user enters a car's specifications manually (make, model, year, mileage, etc.) and the system returns a predicted market price.

Priority: High

- REQ-3.1.1: The system shall provide a form for the user to input car specifications (make, model, year, mileage, fuel type, transmission).
- REQ-3.1.2: The system shall validate all required fields before submission.
- REQ-3.1.3: The system shall send the input data to the prediction model and return an estimated price.
- REQ-3.1.4: The system shall display the predicted price in Egyptian Pounds (EGP).

3.2 Price Prediction by VIN

Description: The user enters a car's Vehicle Identification Number (VIN), and the system automatically retrieves the car's specifications and returns a predicted price, without the user manually entering each field.

Priority: High

- REQ-3.2.1: The system shall provide an input field for the user to enter a VIN.
- REQ-3.2.2: The system shall validate the VIN format before processing.
- REQ-3.2.3: The system shall decode the VIN to retrieve car specifications (make, model, year, and other relevant attributes).
- REQ-3.2.4: The system shall pass the decoded specifications to the prediction model and return an estimated price.
- REQ-3.2.5: The system shall notify the user if the VIN is invalid or cannot be decoded, and shall clearly flag any specification fields that were estimated rather than decoded directly.

3.3 Car Market Dashboard

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Description: An embedded dashboard (built with Tableau) that visualizes trends and insights from the car dataset used to train the prediction model — giving users a broader view of the used car market beyond a single price prediction, based on the project's Exploratory Data Analysis (EDA).

Priority: Medium

- REQ-3.3.1: The system shall display an embedded dashboard showing key market insights (e.g. average price by make/model, price distribution, mileage vs. price trends).
- REQ-3.3.2: The system shall allow the user to filter dashboard data by attributes such as make, model, year, or price range.
- REQ-3.3.3: The system shall present pricing figures in Egyptian Pounds (EGP).
- REQ-3.3.4: The dashboard shall load within the main platform (embedded, not an external redirect).

3.4 Admin Logs Page

Description: A restricted page accessible only to admins, showing a record of all prediction queries made through the system (both manual input and VIN-based).

Priority: Medium

- REQ-3.4.1: The system shall restrict access to the logs page to authorized admin users only (via API key).
- REQ-3.4.2: The system shall log every prediction query, including timestamp, input method (manual/VIN), and success/failure status.
- REQ-3.4.3: The system shall display logs in a readable, sortable format (e.g. table, most recent first).
- REQ-3.4.4: Logged data shall persist independently of the backend server's lifecycle (e.g. stored in an external database), so logs are not lost on server restart.

Non Functional Requirements

- Non-functional requirements describe how well the system performs its functions — covering performance, scalability, security, reliability, and usability.

	Details
Performance	The API shall return a predicted price within 3 seconds for both manual and VIN based requests, and be working 10 hours per day.
Scalability	The system shall be deployable on cloud infrastructure that supports scaling to handle 100 concurrent prediction requests.
Security	The admin logs page shall be accessible only via a valid API key, compared using a constant-time check to prevent timing attacks.

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Reliability	Logged prediction operations shall persist in an external database, surviving server restarts or redeployments.
Usability	The frontend shall present Arabic-language content correctly, including proper right-to-left rendering where Arabic and Latin text are mixed.

System Models

Use Case Model

- The main actors of the system are the User (buyer, seller, or insurance representative) and the Admin. Users can request a price prediction (manually or via VIN) and view the market dashboard; the Admin can additionally view the query logs.

Use Case Tables

- The table below documents the VIN-based prediction use case as an example.

Use Case ID:	UC-01	
Use Case Name:	Predict Car Price by VIN	
Actors:	User	
Pre-conditions:	User has access to the prediction page	
Post-conditions:	Predicted price is displayed to the user and the operation is logged by the system.	
Flow of events:	**User Action**	**System Action**
	1- User enters a VIN, along with mileage, trim, and transmission.	
		2- System decodes the VIN via the NHTSA API and maps it to trained categories.
	3- System merges decoded data with user input and validates it.	

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		4- System runs the prediction model and returns the estimated price and specifications of the car.
Exceptions:	**User Action**	**System Action**
	1- User enters an invalid or unrecognized VIN.	
		2- System cannot resolve make, model, or year from the VIN. 3- System skips the prediction and returns a message specifying which field(s) could not be resolved.
Includes:	Operation logging (the request is recorded for the Admin Logs Page use case).	
Notes and Issues:	1. Fields NHTSA cannot provide (body type, fuel type, drivetrain, engine capacity) are estimated and clearly flagged as estimated in the response, rather than silently guessed. 2. The system could not predict the price of cars (make, model and year) that doesn't include in the training dataset.	

- The table below documents the manual entered specifications based prediction use case as an example.

Use Case ID:	UC-03
Use Case Name:	View Prediction Logs (Admin)

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Actors:	User	
Pre-conditions:	Admin has a valid API key; prediction operations have already been logged in the external database.	
Post-conditions:	Admin views the complete, up-to-date history of prediction queries made through the system.	
Flow of events:	**User Action**	**System Action**
	1- User enters specifications, along Admin navigates to the logs page and provides the API key.	
		2- System validates the API key using a constant-time comparison. 3- System retrieves all logged operations from the external database. 4- System displays the logs in a sortable table (timestamp, input method, success/failure status), most recent first.
Exceptions:	**User Action**	**System Action**
	1- Admin provides an invalid or missing API key.	
		2- System rejects the request and denies access to the logs page..
Includes:	None (this use case does not trigger a prediction; it only reads previously logged data).	

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Notes and Issues:	There is no full login/sign-up system — access is gated by a single shared API key rather than per-user authentication. This is a known limitation flagged for future improvement (see Future Work).

Use Case Tables

- The table below documents the Admin Logs Page feature

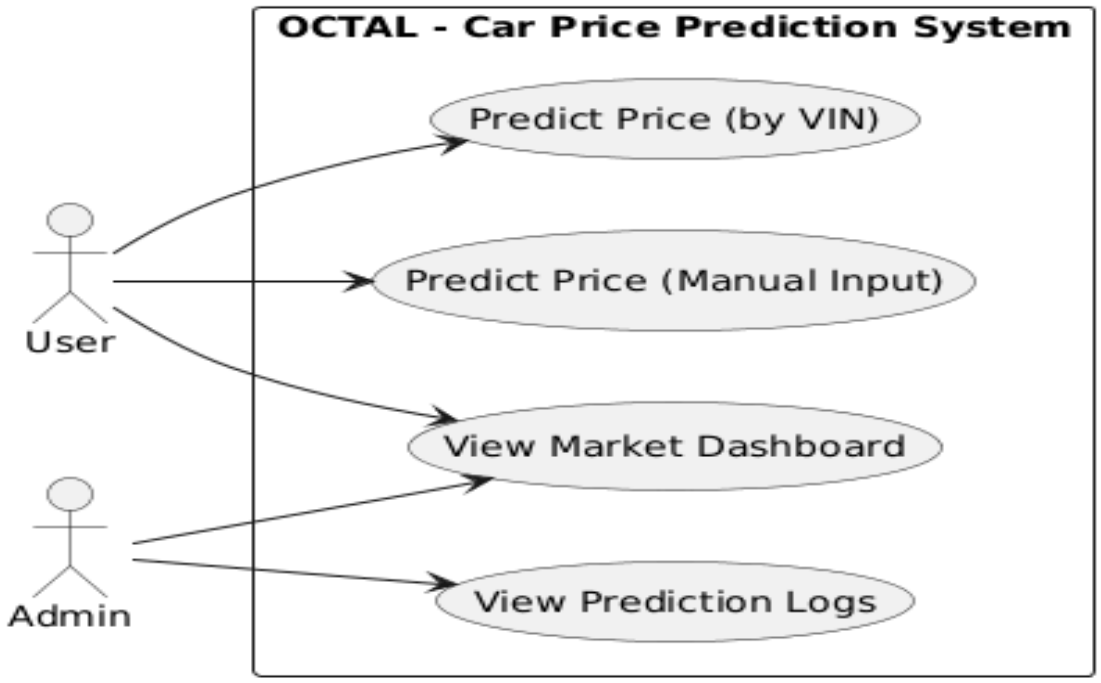
Use Case ID:	UC-03	
Use Case Name:	Predict Car Price by VIN	
Actors:	User	
Pre-conditions:	User has access to the prediction page	
Post-conditions:	Predicted price is displayed to the user and the operation is logged by the system.	
Flow of events:	**User Action**	**System Action**
	1- User enters a VIN, along with mileage, trim, and transmission.	
		2- System decodes the VIN via the NHTSA API and maps it to trained categories.

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	3- System merges decoded data with user input and validates it.	
		4- System runs the prediction model and returns the estimated price and specifications of the car.
Exceptions:	**User Action**	**System Action**
	1- User enters an invalid or unrecognized VIN.	
		2- System cannot resolve make, model, or year from the VIN. 3- System skips the prediction and returns a message specifying which field(s) could not be resolved.
Includes:	Operation logging (the request is recorded for the Admin Logs Page use case).	
Notes and Issues:	3. Fields NHTSA cannot provide (body type, fuel type, drivetrain, engine capacity) are estimated and clearly flagged as estimated in the response, rather than silently guessed. 4. The system could not predict the price of cars (make, model and year) that doesn't include in the training dataset.	

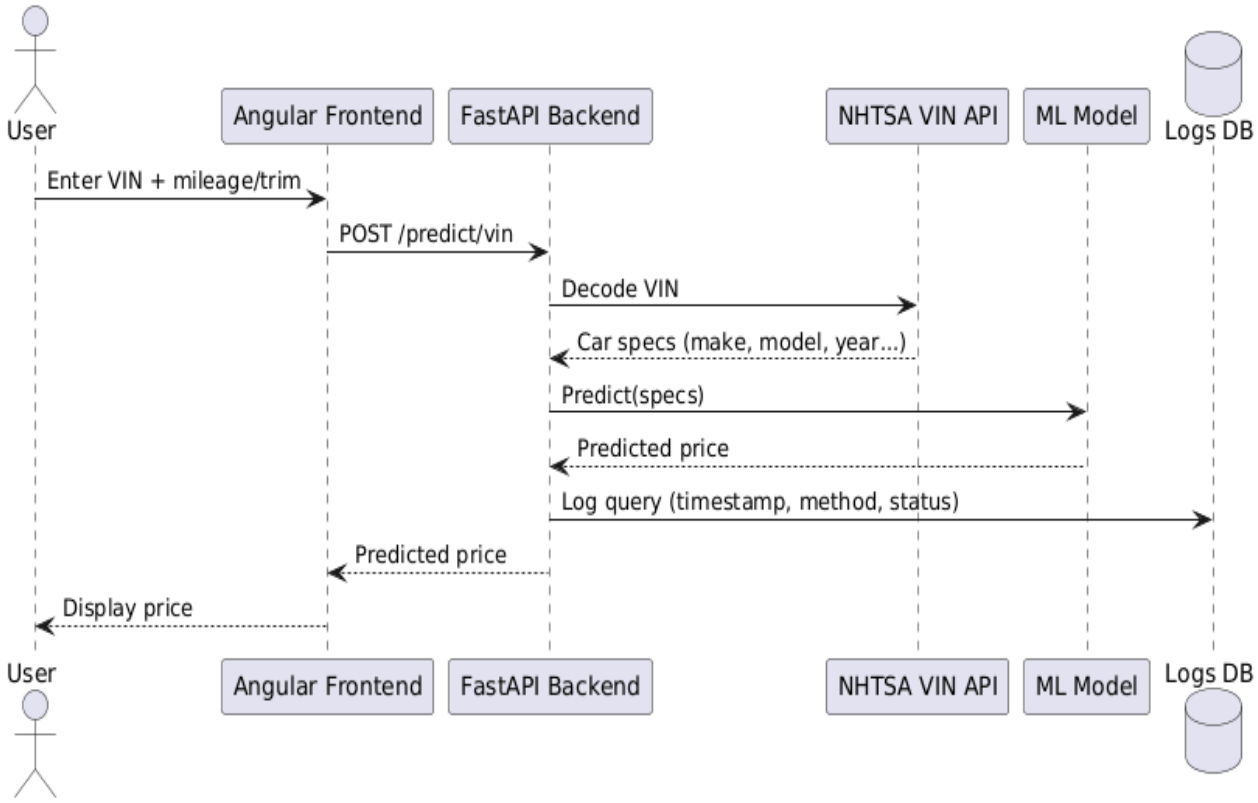
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Use case diagram:



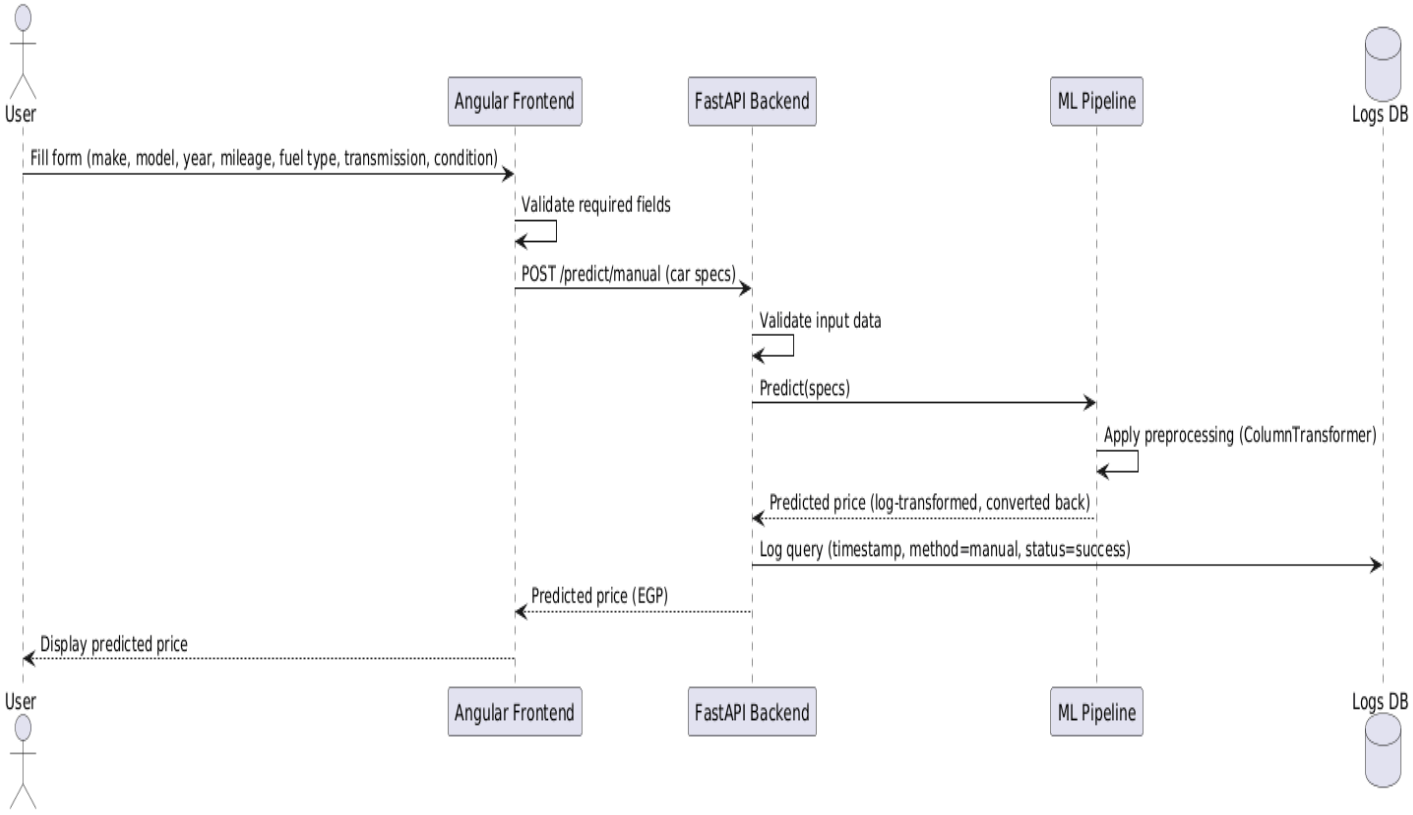
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Sequence diagram for vin:

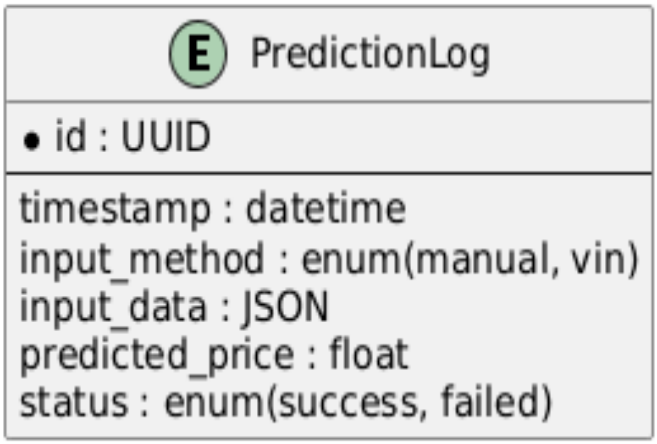


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Sequence diagram for manual input:



ERD Diagram:



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Ownership Report:

Item	Owners
ML-Model, Backend (Fast API) ,Hosting on server.	*Eyad Mohamed ELghonemy*
Front-End With Angular Framework (User Interface For System)	*Abdelrahman Abo ElFadl*
Business Analyst- Collect data- EDA- Tableau Dashboard	*Khalid Yousry Mohamed*